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Digitalization of Bulk Cargo Transportation for Sustainable Port–City Development: Technologies, Governance, and Research Directions

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Abstract: Bulk cargo—dry commodities such as grain, coal, ores, and construction materials, plus liquid bulk such as petroleum products, liquefied gases, and chemicals—is central to urban material supply, yet remains comparatively under-digitalized relative to containerized logistics and is a persistent source of congestion, dust, noise, and emissions at the port–city interface. Dry and liquid bulk involve substantially different handling technologies, risks, and regulatory regimes and are therefore treated as distinct sub-segments rather than as a single homogeneous category. Based on an integrative review of peer-reviewed studies, industry analyses, and policy reports published or first made available online between January 2015 and April 2026, bulk-specific evidence is distinguished from findings derived from broader port and container logistics. Five interacting technology clusters are identified: Internet of Things (IoT) sensing and connectivity; Artificial Intelligence (AI) and predictive analytics; digital twins and simulation; blockchain-enabled documentation; and automation and autonomous systems. The functions and sustainability implications of these clusters are mapped across the economic, environmental, and social dimensions of port–city development. A four-layer conceptual framework—physical assets, data infrastructure, intelligence services, and governance—is proposed, together with testable propositions linking these layers to urban outcomes. Land-use conflict, freight-corridor planning, environmental zoning, and equity in the distribution of port-related environmental burdens are integrated directly into the framework. The trade-offs associated with digitalization—including energy use, cybersecurity exposure, labor transition, technological dependence and interoperability failures, digital exclusion of smaller operators, and rebound effects—are critically assessed. A stakeholder-differentiated roadmap and research agenda are also proposed to support future bulk-specific empirical validation.

Keywords: Bulk cargo transportation; Digitalization; Digital twin; Smart port; Port–city interface; Urban freight; Sustainable urban development

1 Introduction

Cities depend on continuous flows of bulk materials. Construction aggregates, cement, grain, fertilizers, coal, ores, and liquid fuels move through ports, rail yards, inland waterways, and road corridors that often sit inside dense urban fabric. Bulk commodities account for a substantial share of global seaborne trade by volume [1]. Even so, research on bulk-terminal operations remains less developed than the corresponding literature on container terminals [2, 3]. This gap has direct urban policy implications: bulk handling and haulage contribute to heavy-vehicle traffic, fugitive dust, noise, and emissions at the port–city interface [4–7], while inefficient bulk chains can increase truck movements and waiting times.

At the same time, digital transformation is reshaping maritime and urban freight systems. We use "digitalization" throughout to denote the broader process of embedding digital technologies into physical operations, and "digital transformation" to denote the accompanying organizational and governance change; the two terms are used consistently in this sense for the remainder of the paper. International organizations have urged ports to progress from paper-based procedures toward port management systems and, ultimately, smart port models built on Artificial Intelligence (AI), advanced analytics, Internet of Things (IoT), 5G connectivity, autonomous systems, digital twinning, and distributed ledger technologies [8]. Research on smart cities similarly indicates that digital technologies and data analytics can materially improve the sustainability of urban freight when integrated with planning and governance rather than deployed in isolation [9, 10]. Digital twins in particular are emerging as instruments that connect port operations with city-scale planning concerns [11–13].

This paper addresses two research questions (RQs). RQ1: Which digital and innovative technologies documented in the recent literature are applicable to bulk cargo transportation—distinguishing dry from liquid bulk where the evidence permits—and through which operational mechanisms do they act? RQ2: How can these technologies be linked to the economic, environmental, and social dimensions of sustainable urban development at the port–city interface, and under what governance conditions do these links hold? We address these questions through an integrative literature review and conceptual synthesis. The paper makes four contributions: it consolidates dispersed evidence on the comparatively underexplored bulk segment; it distinguishes bulk-specific findings from evidence derived from broader port and logistics contexts; it maps major technology clusters to economic, environmental, and social sustainability outcomes; and it proposes a layered conceptual framework, supported by testable propositions, that integrates urban planning and governance concerns.

The rest of the paper is structured as follows. Section 2 reviews the relevant literature and defines the scope and characteristics of bulk cargo transportation. Section 3 describes the review methodology. Section 4 presents the synthesis of findings, including the technology-to-sustainability mapping and the proposed conceptual framework. Section 5 discusses practical implications, trade-offs, and the stakeholder roadmap. Section 6 presents the study limitations and future research directions, and Section 7 concludes the paper.

2 Literature Review

2.1 Digitalization of Ports and Freight Transport

The digital transformation of ports has been described as an evolutionary trajectory that moves from isolated electronic data interchange toward integrated port community systems and, eventually, smart ports in which emerging technologies improve performance, competitiveness, and environmental sustainability [8]. Efficient cargo flows also depend on the coordination of transportation, forwarding, and warehousing services within the broader logistics system [14]. Studies of individual ports provide early quantitative evidence: analyses of digital transformation programs report significant positive effects on cargo and container throughput, with long-term effects exceeding short-term ones and regional economic conditions moderating the impact [15]. Reviews of digitalization and automation in shipping likewise document the diffusion of AI, blockchain, and automation across terminal operations and vessel management [16, 17]. The literature consistently finds that the value of digitalization lies less in creating new information systems than in linking existing databases and stakeholders through interoperable platforms and shared data standards [8]. This body of evidence, however, is drawn largely from container terminals and general port operations and should therefore be distinguished from bulk-specific evidence unless a study explicitly addresses bulk cargo.

2.2 The Specificity of Bulk Cargo Chains

Bulk cargo chains differ from containerized logistics in ways that shape technology adoption. Cargo is unpackaged and flow-like, so quality, moisture, temperature, and contamination must be monitored continuously rather than at discrete handover points. Handling relies on specialized assets—conveyors, silos, stockyards, and loaders—whose utilization, wear, and maintenance dominate terminal economics; predictive-maintenance research on conveyor systems, the workhorse asset of bulk terminals, has grown into a distinct literature documenting condition monitoring via vibration, thermal, and current sensing combined with machine learning to estimate remaining useful life [18]. Documentation has historically been paper-intensive. Many bulk terminals are also legacy facilities located close to residential districts, which amplifies dust, noise, and traffic externalities. Dedicated environmental-monitoring studies at industrial port areas identify unconfined bulk-solid handling, rather than shipping itself, as the dominant source of particulate matter with an aerodynamic diameter of 10 µm or less (PM10) emissions and recommend PM10 rather than particulate matter with an aerodynamic diameter of 2.5 µm or less (PM2.5) as the priority metric for bulk-terminal monitoring [5]. Complementary evidence from dry-bulk ports links fine-dust exposure directly to cargo-handling operations and frames dust management as a sustainable supply-chain concern specific to bulk terminals [4]. The literature on smart ports acknowledges that digitalization and automation can reduce processing time and cost across unloading, transportation, storage, and management of different cargo types, and can remove workers from dangerous tasks through unmanned equipment [16]. Recent bulk-specific digital-twin research demonstrates a working digital twin for a dry-bulk biomass terminal that integrates Building Information Modelling with sensor and synthetic data to assess resilience and sustainability where high-resolution IoT infrastructure is still limited—explicitly noting that, despite abundant container-terminal digital-twin research, bulk-terminal digital twins remain scarce [2]. Studies focusing specifically on bulk segments nevertheless remain scarce relative to container-oriented research, a gap that this review seeks to address by giving particular attention to bulk-specific evidence.

2.3 Boundaries of Bulk Cargo: Dry and Liquid Bulk Cargo Types

Dry bulk (e.g., grain, coal, ores, cement, construction aggregates) and liquid bulk (e.g., crude oil, refined petroleum products, liquefied gases, and bulk chemicals) are not a homogeneous logistics category. They differ in handling technology, dominant safety hazards, environmental externalities, and regulatory regimes, and the digital technologies most relevant to each sub-segment differ accordingly. Table 1 summarizes these differences and indicates which technology clusters are best supported by the evidence for each cargo type; this review's scope covers both sub-segments but does not assume that findings transfer automatically between them.

Table 1. Dry bulk versus liquid bulk cargo: handling, risk, and digitalization relevance

Dimension	Dry Bulk (Grain, Coal, Ores, Cement, Aggregates)	Liquid Bulk (Petroleum, LNG/LPG, Chemicals)
Primary handling assets	Conveyors, stackers/reclaimers, silos, stockyards, grabs/hoppers	Pipelines, storage tanks, pumps, manifolds, jetties
Dominant environmental externality	Fugitive dust, PM10, noise	Vapor emissions, spill risk, groundwater/marine contamination risk
Dominant safety hazard	Self-heating/combustion of organic bulk, dust explosion, engulfment	Fire, explosion, toxic release, pipeline/tank failure
Regulatory emphasis	Air-quality and dust-control permitting; food/feed safety for agribulk	MARPOL/hazmat, process-safety, and pollution-prevention regimes
Best-supported technology clusters in current evidence	IoT dust/air-quality sensing [4, 5]; conveyor predictive maintenance [18]; dry-bulk digital twins [2]; grain-chain blockchain provenance [19]	Blockchain-enabled trade documentation and secure transaction management for oil logistics [20]
Documentation intensity	High (quality, origin, phytosanitary, customs certificates)	Very high (quality, custody transfer, hazmat, customs, insurance)

Note: LNG = liquefied natural gas; LPG = liquefied petroleum gas; PM10 = particulate matter with an aerodynamic diameter of 10 µm or less; MARPOL = International Convention for the Prevention of Pollution from Ships; hazmat = hazardous materials; IoT = Internet of Things.

2.4 Urban Freight, Smart Cities, and Sustainability

A complementary research stream connects freight digitalization to urban sustainability. Work on smart cities and urban freight logistics shows that sensor systems, IoT devices, and data analytics can support decision-making by planners, operators, and residents, and frames city logistics as a central component of sustainable urban development [9]. Reviews of digital twin applications in urban logistics find that virtual replicas of transport networks can support the simulation of traffic flows, delivery operations, and environmental impacts before implementation, with applications reported in cities such as Stockholm, Singapore, and Helsinki [11]. Studies of port cities with mixed cargo profiles further show that digital interfaces, including shared traffic-sensor data, video analytics, and city data models, can support traffic management and the simulation of development scenarios around port areas [13]. Microsimulation studies of heavy-vehicle flow near city-centre terminals quantify how terminal-generated truck traffic interacts with urban road networks and propose truck-arrival management as a mitigation lever [6]; complementary port–city corridor studies similarly highlight the value of dry ports and capacity-expansion planning in relieving urban-interface congestion [7]. Recent studies extend the digital twin concept to the port–city interface and suggest that port digital twins can support more integrated and sustainable port–city development [12]. Systematic reviews of advanced digital technologies in multimodal logistics indicate that Industry 4.0 technologies can contribute to more efficient and intelligent logistics systems and improved sustainability performance [21], while studies of port digitalization identify interoperability, system integration, and fragmented data environments as important implementation challenges [22, 23]. Policy-oriented sources further indicate that developing cities face particular institutional complexity in steering urban freight toward more sustainable pathways [24].

2.5 Scope of This Review

This review covers evidence from bulk cargo transportation as well as relevant studies from the broader port, maritime, and urban logistics literature. Particular attention is given to distinguishing findings derived directly from dry or liquid bulk cargo applications from those extrapolated from general port or container logistics. Cross-cutting studies on smart-city systems, sustainability, energy use, and digitalization are also considered where they provide relevant context for the port–city interface. This distinction is maintained throughout the review to avoid treating findings from broader logistics settings as established evidence for bulk cargo applications.

3 Methodology

This study employs an integrative literature review combined with conceptual framework development. This approach is appropriate for synthesizing heterogeneous literature across maritime economics, logistics engineering, environmental monitoring, smart-city research, and urban planning, with the aim of developing an integrated understanding of digitalization in bulk cargo transportation rather than conducting a statistical meta-analysis or formal systematic review.

Relevant literature published or first made available online between January 2015 and April 2026 was identified from major academic databases and repositories, including Scopus, Web of Science, ScienceDirect, SpringerLink, MDPI, Taylor & Francis, and arXiv. Policy and institutional reports from the World Bank, the United Nations Economic and Social Commission for Asia and the Pacific (ESCAP), UN Trade and Development (UNCTAD), Deutsche Gesellschaft für Internationale Zusammenarbeit (GIZ), and the U.S. Government Accountability Office were also considered where relevant. Search terms combined concepts related to bulk cargo and port logistics, digital technologies, and sustainable urban development, including "bulk cargo," "bulk terminal," "dry bulk," "liquid bulk," "port," "urban freight," "digitalization," "digital transformation," "IoT," "AI," "digital twin," "blockchain," "automation," "sustainability," "smart city," "air quality," "dust," and "congestion."

Sources were included when they either (i) addressed at least one digital technology in a freight, port, or urban logistics context and discussed relevant operational or sustainability implications; (ii) provided cross-cutting evidence on the energy or environmental implications of digitalization relevant to the sustainability assessment; or (iii) provided authoritative contextual evidence on freight and logistics systems, maritime trade, or policy conditions necessary to frame the review. Selected industry sources were retained where they provided relevant practical evidence not sufficiently documented in the peer-reviewed literature.

The selected literature was synthesized thematically according to the major technology clusters identified in the review and their economic, environmental, and social implications for sustainable port–city development. Particular attention was given to distinguishing evidence directly related to bulk cargo from evidence derived from general port or container logistics. This synthesis informed the technology-to-sustainability mapping and the conceptual framework presented in the subsequent sections.

4 Results

The reviewed literature reveals five major technology clusters relevant to the digitalization of bulk cargo transportation: IoT sensing and connectivity, AI and predictive analytics, digital twins and simulation, blockchain and electronic documentation, and automation and autonomous systems. The following sections synthesize the main applications of these technologies and their implications for sustainable port–city development.

4.1 Internet of Things Sensing and Connectivity

IoT is the sensory foundation of digital bulk logistics. In port and logistics contexts relevant to bulk operations, documented applications include continuous monitoring of cargo condition (moisture, temperature, weight) in silos, holds, and tanks; telematics for truck and rail fleets; smart weighbridges and automated gate systems; and environmental sensing of dust, noise, and air quality around terminals [10, 16]. Dedicated port air-quality monitoring studies confirm that solid bulk handling in unconfined facilities is a primary local source of coarse particulate matter and recommend continuous PM10 sensor networks, rather than periodic sampling, as the appropriate monitoring strategy for bulk terminals specifically [5]; dry-bulk port studies extend this to fine-dust exposure management as a distinct sustainable-supply-chain concern [4]. Real-time data from sensors and tracking devices enable congestion forecasting and coordination across port stakeholders [8]. For urban sustainability, IoT matters in two ways: it turns previously invisible externalities such as fugitive dust into measurable, manageable quantities—directly supporting the community-transparency function discussed in Section 4.7—and it supplies the data streams on which all higher-level analytics depend [5]. Current bulk-specific evidence is strongest for dust and air-quality sensing [4, 5], whereas evidence concerning telematics and gate systems is drawn mainly from the broader port and logistics literature [8, 16].

4.2 Artificial Intelligence and Predictive Analytics

AI and machine learning act on IoT data to optimize decisions. Applications reported in the port and urban logistics literature include berth and yard allocation, vessel and truck arrival prediction, dynamic routing, demand forecasting, and predictive maintenance of handling equipment [8, 16, 21]. In bulk chains, predictive maintenance is particularly consequential because conveyor, stacker-reclaimer, or shiploader failure halts the entire flow; a systematic review of digital predictive-maintenance technologies for conveyor systems documents condition monitoring via vibration, thermal, and current sensing combined with cloud/edge computing and machine learning to estimate remaining useful life, which is directly transferable to bulk-terminal conveyor networks [18]. IoT-enabled predictive analytics in smart city logistics have also been shown to improve delivery and traffic management when embedded in feedback-based control architectures [10]. The sustainability effects operate mainly through utilization: better forecasting and scheduling reduce vessel idle time at anchor, truck queuing at gates, and empty runs, lowering fuel consumption and emissions per tonne transported and easing heavy-traffic pressure on urban roads [9, 21]. Bulk-specific evidence is available for conveyor predictive maintenance [18], while berth allocation, yard management, and routing applications are supported mainly by the broader port and logistics literature [8, 16, 21].

4.3 Digital Twins and Simulation

Digital twins—virtual, data-fed replicas of physical systems—have moved from manufacturing into urban transport and port management [11, 12]. City-scale twins are used to simulate transportation networks, freight flows, and environmental impacts in real time, so that strategies can be tested before real-world implementation [10, 11]. Port digital twins are increasingly discussed as instruments of integrated and sustainable port–city development [12]. Directly within the bulk segment, a recently reported digital twin for a dry-bulk biomass terminal integrates Building Information Modelling and sensor/synthetic data to assess resilience and sustainability outcomes, explicitly addressing the scarcity of bulk-terminal (as opposed to container-terminal) digital-twin research [2]. Studies of mixed-cargo port cities similarly report that shared digital-twin and traffic-sensor platforms can support the modelling of port-area development scenarios and communication with municipal stakeholders [13]. For bulk transportation, digital twins permit scenario analysis of terminal expansion, stockyard configuration, corridor routing, and modal shift, together with advance assessment of noise and emission footprints on adjacent neighborhoods. This ability to test options before committing to them aligns digital twins closely with sustainable urban planning practice, including participatory evaluation of alternatives with municipal stakeholders [11–13]. Direct bulk-specific evidence remains limited, although a notable dry-bulk terminal digital-twin application has been reported [2]. Broader port–city and digital-twin applications are primarily supported by general port and urban logistics studies [11–13].

4.4 Blockchain and Electronic Documentation

Distributed ledger technologies address the transactional layer of bulk logistics. Blockchain has been explored for secure and tamper-evident recording of transactions and shipping documentation [19, 20], and can support electronic documentation, commodity provenance, and trusted data exchange within port and logistics systems [8, 19, 20]. Bulk-specific applications now exist on both sides of the dry/liquid divide: a blockchain framework for oil-port logistics targets the high-volume, highly certified documentation (custom declarations, bills of lading, custody-transfer records) characteristic of liquid-bulk trade, using private-blockchain smart contracts to ensure tamper-evident compliance among trading parties [20]; and a grain-exporters' blockchain business network links producers, cooperatives, warehouses, and freight forwarders in a shared ledger for quality-assurance and provenance tracking across a multi-stage dry-bulk commingling chain [19]. Bulk trades in general involve high-value single shipments with extensive certification (quality, origin, phytosanitary, customs), so digitized and cryptographically verifiable documentation can shorten clearance times and reduce vessel and truck dwell in urban port areas. The literature cautions, however, that the benefit of such systems depends on multi-stakeholder adoption and standards alignment rather than on the technology itself [22, 23]. Bulk-specific applications have been reported for oil-port and grain-chain documentation [19, 20], while broader blockchain applications are supported by the general port and logistics literature [8].

4.5 Automation and Autonomous Systems

Automation spans terminal equipment (automated stacker–reclaimers, cranes, and gate systems), autonomous or remotely operated vessels, and autonomous trucks on fixed port–hinterland shuttles [16]. Industry reporting on bulk-terminal automation specifically documents production-efficiency gains of up to 50% and reduced vessel turnaround at automated dry-bulk facilities, attributing these to optimized crane and conveyor operation together with AI-enabled inventory and stockpile tracking of cargo volume and moisture content [25]. A national assessment of port automation similarly finds that automation can improve worker safety by separating personnel from moving equipment and can reduce emissions by improving efficiency, while also reporting mixed effects on throughput and workforce size that vary by terminal and equipment type [17]—evidence drawn predominantly from container terminals but relevant by extension to bulk facilities considering automation. Combined with electrification, automation can smooth equipment duty cycles and enable night-time or off-peak operations that relieve daytime urban congestion. The same literature notes significant labor-transition and cybersecurity implications, which we return to in Section 5.2 [16, 17, 21]. Bulk-specific evidence for automation is currently limited [25], while much of the available evidence is derived from broader port and container-terminal applications [16, 17, 21].

4.6 Cross-Mapping of Technologies and Sustainability Outcomes

Table 2 brings the reviewed evidence together by mapping each technology cluster to its principal operational functions, sustainability outcomes, and supporting evidence.

Table 2. Technology clusters in bulk cargo transportation, sustainability outcomes, and supporting evidence

Technology Cluster	Principal Functions in Bulk Chains	Economic / Environmental / Social Outcomes	Evidence Basis (Citations)
IoT sensing and connectivity	Cargo condition monitoring; fleet telematics; smart weighbridges; dust/noise/air-quality sensing	Economic: reduced cargo losses and demurrage; Environmental: detection/management of fugitive dust and air-quality impacts; Social: occupational safety; transparency to affected communities	Bulk-specific (dust/air-quality sensing) [4, 5]; general port (telematics/gates) [8, 16]
AI and predictive analytics	Berth/yard allocation; arrival prediction; routing; demand forecasting; predictive maintenance	Economic: higher asset utilization, shorter turnaround; Environmental: lower fuel use and CO_2 per tonne; Social: reduced heavy-vehicle pressure on urban roads	Bulk-specific (conveyor predictive maintenance) [18]; general port (berth/yard/routing) [8, 16, 21]
Digital twins and simulation	Virtual replicas of terminals, corridors, port–city interface; scenario testing	Economic: de-risked investment, capacity optimization; Environmental: advance assessment of noise/emission footprints; Social: participatory, evidence-based planning with city stakeholders	Bulk-specific (dry-bulk terminal twin) [2]; general port, extrapolated (port–city twins) [11–13]
Blockchain and e-documentation	Electronic bills of lading; commodity provenance; port community systems; single windows	Economic: faster clearance, less paperwork and fraud; Environmental: shorter vessel/truck dwell and associated emissions; Social: trust and accountability among supply chain actors	Bulk-specific (oil-port, grain-chain) [19, 20]; general port (community systems) [8]
Automation and autonomous systems	Automated handling equipment; autonomous shuttles; gate automation; remote operation	Economic: extended operating hours, labor productivity; Environmental: electrified, optimized equipment cycles; off-peak operations; Social: workers removed from hazardous zones; labor transition to be managed	Bulk-specific (automated bulk terminals) [25]; general port, extrapolated [16, 17, 21]

Note: AI = Artificial Intelligence; CO_2 = carbon dioxide; IoT = Internet of Things.

4.7 Conceptual Framework and Theoretical Propositions

Building on this mapping, we propose a four-layer conceptual framework that links digitalization of bulk cargo transportation to sustainable urban development. Layer 1 (physical assets) comprises vessels, vehicles, terminals, and handling equipment. Layer 2 (data infrastructure) comprises IoT sensors, connectivity (including 5G), data standards, and cybersecurity, which make the physical layer observable. Layer 3 (intelligence services) comprises AI analytics, digital twins, and optimization services that convert data into decisions. Layer 4 (governance) comprises port community systems, blockchain-based trust mechanisms, regulation, and formal port–city collaboration, which align decisions across stakeholders. Sustainability outcomes emerge from the interaction of all four layers: efficient and resilient urban material supply (economic), lower emissions, dust, and noise (environmental), and safer work and more livable port districts (social). Figure 1 illustrates the four layers and the proposed relationships among physical assets, data infrastructure, intelligence services, governance, and sustainability outcomes.

Based on the reviewed literature and the proposed framework, five conceptual propositions are suggested for future empirical investigation:

P1 (Data infrastructure as mediator). Layer 2 mediates the relationship between Layer 1 and Layer 3. Independent variable: the degree of instrumentation of Layer-1 physical assets. Dependent variable: the sustainability contribution realized through Layer-3 AI and digital-twin services. Mediator: the quality and standardization of data provided through Layer 2. AI and digital-twin services can act effectively on physical assets only to the extent that these assets are adequately instrumented and their data are available, reliable, and standardized; where sensor coverage or data quality is poor, intelligence-layer investment is likely to yield limited sustainability benefits. Expected direction: positive, with greater sensor coverage, data quality, and data-standard maturity expected to strengthen the contribution of Layer-3 services. Unit of analysis: individual bulk terminal or berth. Candidate indicators: proportion of handling assets under continuous condition monitoring; data-standard conformance rate; and realized versus nominal analytics accuracy. This proposition is consistent with, though not directly tested by, evidence that bulk digital twins have had to rely on synthetic data where IoT infrastructure was limited [2].

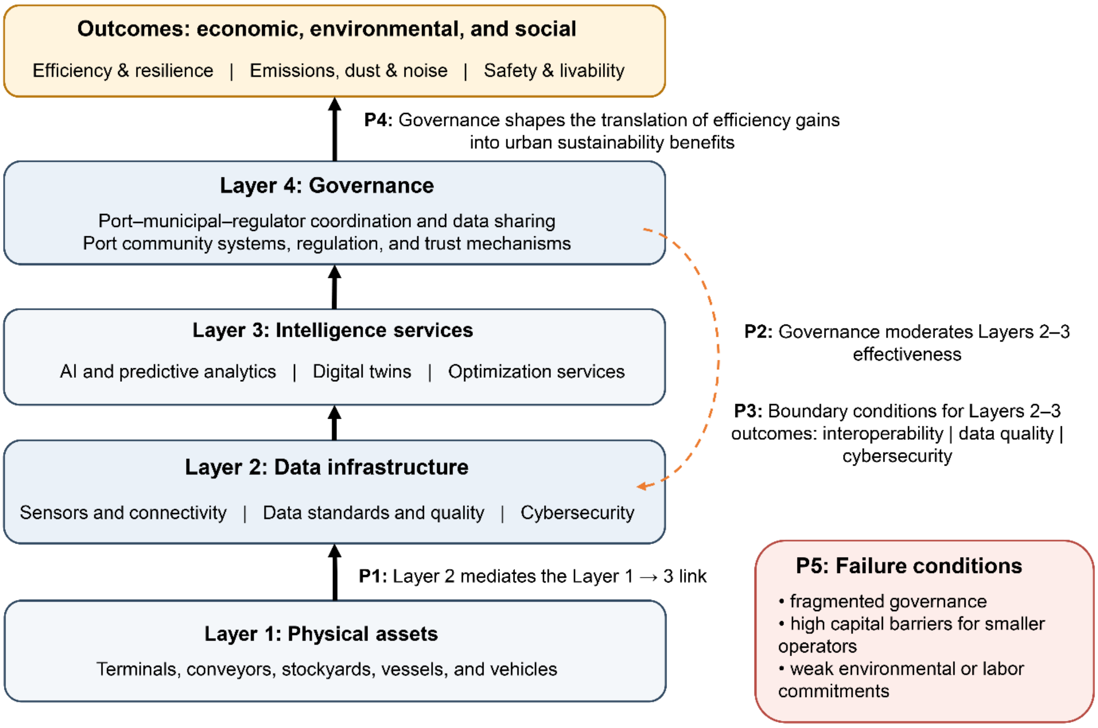

Figure 1. Four-layer conceptual framework linking digitalization of bulk cargo transportation to sustainable port–city development

Note: AI = Artificial Intelligence.

P2 (Governance as moderator). Layer 4 moderates the effectiveness of Layers 2–3. Independent variable: sensing and analytics investment across Layers 2–3. Dependent variable: realized sustainability gains in economic, environmental, or social terms. Moderator: the presence and formality of Layer-4 data-sharing, coordination, and interoperability arrangements among ports, operators, and municipalities. Expected direction: positive interaction, whereby the same level of sensing and analytics investment is expected to produce greater sustainability gains where formal governance and data-sharing arrangements are in place. Unit of analysis: port authority or port–municipality dyad. Candidate indicators: existence of a formal data-sharing charter; number of active inter-agency data-exchange agreements; and time from sensor deployment to shared operational use. This proposition follows from, but extends beyond, findings in the broader logistics literature that interoperability and multi-stakeholder adoption influence the realization of digitalization benefits [21–23].

P3 (Interoperability, data quality, and cybersecurity as boundary conditions). Independent variable: deployment of Layers 2–3 digital technologies. Dependent variable: net sustainability outcomes rather than gross operational efficiency alone. Boundary conditions: interoperability across systems, data quality, and cybersecurity capacity. Expected relationship: the sustainability benefits associated with Layers 2–3 are expected to be stronger where these conditions are adequately addressed; weaknesses in any of them may reduce the expected benefits or create additional operational and security risks. Unit of analysis: terminal digital system, including an individual platform or integrated digital stack. Candidate indicators: system uptime and interoperability-failure rate; data-quality audit pass rate; and recorded cybersecurity incidents per year.

P4 (Economic-to-urban translation is not automatic). Independent variable: economic efficiency gains generated through Layers 1–3, such as higher asset utilization and shorter turnaround times. Dependent variable: urban environmental or social benefits, such as reduced dust, emissions, or heavy-vehicle pressure. Moderator: the extent to which Layer-4 governance channels capacity or cost savings toward externality reduction, for example through dust suppression, cleaner operations, or off-peak scheduling, rather than toward freight-volume growth alone. Expected direction: the translation of economic efficiency into urban sustainability benefits is expected to be stronger where appropriate governance mechanisms are present; otherwise, efficiency gains may enable greater freight movement without a corresponding reduction in externalities. Unit of analysis: terminal–city corridor over a defined multi-year period. Candidate indicators: freight-volume growth relative to efficiency gains; and externality intensity, such as dust or emissions per tonne moved, before and after digitalization.

P5 (Conditions for failure). Independent variables: (a) the presence or fragmentation of Layer-4 governance across port, municipal, and regulatory authorities; (b) the capital intensity of Layers 2–3 adoption relative to operator size; and (c) whether efficiency gains are accompanied by environmental or labor commitments. Dependent variable: net urban sustainability gain from digitalization. Expected relationship: digitalization is less likely to generate net urban sustainability benefits where governance is absent or fragmented, where high capital requirements exclude smaller bulk operators from Layers 2–3, or where efficiency gains are not accompanied by appropriate environmental and labor commitments. Unit of analysis: port or terminal-operator population within a region. Candidate indicators: share of small or independent operators with access to Layers 2–3 technologies; and existence of formal environmental or labor commitments associated with automation and digitalization programs.

Previous studies provide general support for the importance of interoperability and multi-stakeholder adoption [21–23]. However, P1–P5 have not yet been directly tested in a bulk cargo setting and are therefore proposed for future empirical validation rather than presented as demonstrated relationships.

5 Discussion

5.1 Implications for Port Authorities, Terminal Operators, and Municipal Planners

For port authorities and terminal operators, the reviewed evidence suggests a staged digitalization pathway for bulk facilities: instrument assets and flows (Layer 2), deploy analytics and twin-based planning (Layer 3), and build formal data-sharing arrangements with municipal and supply chain partners (Layer 4). This echoes the maturity trajectory recommended for ports generally—from digitized procedures through port management systems to smart port models [8]—but adapts it to bulk-specific priorities such as condition monitoring, dust management, and predictive maintenance of conveyors and stockyards [4, 5, 18].

For municipal planners, these findings also have implications for urban development. Land-use conflict between bulk terminals and adjacent residential districts is not simply a consequence of digitalization but a broader planning issue that the framework's Layer 4 (governance) can help address. Shared air-quality sensor networks (Layer 2) can provide municipal planners with real-time dust and emissions data used by terminal operators, thereby supporting environmental monitoring and evidence-based planning around bulk facilities [5, 13]. Heavy-freight corridor planning and truck-access time-window design may also benefit from AI-based arrival prediction and digital-twin capabilities, which allow planners to assess potential congestion effects before implementation, as demonstrated in studies of the port–city interface [6, 7]. Port–city digital twins can further support infrastructure investment and scenario analysis by enabling joint evaluation of alternative corridor or terminal-expansion scenarios by port authorities, municipal planners, transport agencies, and environmental regulators [12, 13]. Finally, social equity in the distribution of port-related environmental burdens remains an important governance consideration. Digitalization alone cannot ensure an equitable distribution of environmental benefits and burdens; therefore, attention should be given to the distribution of monitoring, mitigation measures, and stakeholder consultation across affected communities.

5.2 Risks, Trade-Offs, and Limits of Digitalization

A rigorous sustainability review must weigh digitalization's costs alongside its benefits rather than emphasizing benefits disproportionately. We address the principal trade-offs here.

(1) Energy use and rebound effects: the sensors, connectivity, cloud/edge computing, AI models, and blockchain infrastructure that underlie Layers 2–3 are themselves energy-consuming. Existing research highlights the environmental implications of supply-chain digitalization [26] and the growing energy demand associated with AI, data centers, and expanding digital infrastructure [27]. In bulk logistics, these energy requirements may partly offset the environmental benefits expected from digital efficiency improvements. Moreover, at the operational level, efficiency gains that lower the cost per tonne moved could potentially encourage greater freight volumes, thereby reducing the net sustainability benefits unless appropriate governance measures are in place.

(2) Cybersecurity and operational vulnerability: connectivity that enables condition monitoring and blockchain documentation also expands the attack surface of port and terminal systems, and port systems have already become targets of cyberattacks [8, 16]. For bulk terminals, an attack on conveyor or stockyard control systems carries direct safety implications given the self-heating and dust-explosion hazards associated with bulk-material handling.

(3) Surveillance and data-privacy concerns: environmental and traffic sensor networks that serve legitimate community-transparency functions also generate granular data on truck movements, worker activity, and, where residential areas are monitored for dust or noise compliance, on nearby communities; governance arrangements for who holds and can access this data remain underdeveloped in the reviewed literature.

(4) Labor displacement and workforce transition: automation removes workers from hazardous tasks but also displaces roles, and national assessments of port automation report mixed workforce effects that vary by terminal and equipment type rather than uniform improvement [17]; the labor-transition governance this implies is a Layer 4 requirement, not an automatic byproduct of automating Layers 1–3.

(5) Technological dependence and interoperability failures: heterogeneous and legacy information systems can create significant interoperability challenges in port digitalization, particularly where data formats and digital platforms are fragmented across different stakeholders [22, 23]. Unresolved interoperability problems may result in parallel, non-communicating systems and increase implementation costs.

(6) Digital exclusion of smaller operators and unequal capacity for adoption: the capital intensity of Layers 2–3 favors large terminal operators and ports in wealthier regions, and developing-city urban freight research documents particular institutional complexity in steering freight systems toward sustainable pathways under resource constraints [24]; without deliberate governance attention, digitalization risks widening rather than narrowing the gap between well-resourced and smaller or developing-city bulk operators.

These trade-offs are not peripheral caveats; they represent important conditions that may limit the sustainability benefits of digitalization and should be treated as design considerations from the outset rather than as risks to be managed only after deployment.

5.3 Operational Roadmap by Stakeholder

Table 3 sets out illustrative short-, medium-, and long-term actions for the principal stakeholder groups identified in the reviewed literature. For each stakeholder, the table also presents the supporting evidence, a measurable performance indicator, and the principal implementation barrier, while the long-term column identifies relevant actions or prerequisites. This roadmap is offered as a structuring device for practice and future case-study research rather than as a validated implementation plan.

Table 3. Illustrative digitalization roadmap by stakeholder, with supporting evidence, indicators, and implementation barriers

Stakeholder	Short-Term Action	Medium-Term Action	Long-Term Action / Prerequisite	Supporting Evidence	Performance Indicator	Principal Barrier
Port authorities	Inventory legacy assets; pilot IoT condition and dust sensing on one bulk terminal	Standardize data formats across terminals; commission a bulk-terminal digital twin	Formal port–city data-sharing charter; sustained funding line for Layer 2–3 upgrades	Extrapolated from general-port digitalization maturity models [8]; bulk digital-twin precedent [2]	Share of bulk assets under continuous condition monitoring	Capital cost of retrofitting legacy conveyor/ stockyard assets
Municipal governments	Request access to existing port air-quality/ traffic sensor feeds	Co-develop environmental zoning and truck-access time windows using shared twin/simulation outputs	Joint port–city planning body with standing data-access rights (Layer 4)	Extrapolated from port–city digital-twin and traffic-microsimulation studies [6, 7, 12, 13]	Number of zoning or access decisions supported by shared sensor/twin data	Absence of a legal basis for routine port–municipality data sharing
Bulk terminal operators	Retrofit conveyors/ stockyards with condition-monitoring sensors	Adopt blockchain or equivalent e-documentation for high-certification cargo	Full instrumentation of Layers 1–2; workforce retraining program tied to automation rollout	Bulk-specific conveyor predictive-maintenance and blockchain evidence [18–20]	Unplanned conveyor downtime (hours per year)	Interoperability and integration failures across legacy and new systems [22, 23]
Logistics and trucking companies	Enroll in terminal appointment/gate systems where available	Integrate telematics with terminal arrival-prediction systems	Fleet-level data-sharing agreements supporting corridor-level congestion management	Extrapolated from general-port arrival-prediction and gate-automation evidence [8, 16, 21]	Average truck dwell/queue time at terminal gates	Fragmented, non-interoperable appointment systems across terminals

Continued

Environmental regulators	Define PM10-based monitoring and reporting standards for bulk terminals	Require public dashboards for terminal-adjacent air-quality data	Harmonized cross-jurisdiction standards for port–city environmental reporting	Bulk-specific dust/PM10 monitoring evidence [4, 5]	Continuous PM10 sensor coverage at bulk-handling points	Absence of harmonized bulk-terminal air-quality reporting standards across jurisdictions
Labor organizations	Establish consultation rights over automation rollout plans	Co-design retraining pathways for displaced roles	Standing labor-transition governance body (Layer 4) with binding review authority	Extrapolated from national port-automation workforce assessment [17]	Share of displaced workers entering funded retraining pathways	Absence of binding consultation or retraining commitments in automation planning
Community stakeholders	Request access to existing environmental sensor dashboards	Participate in digital-twin-based scenario reviews for terminal expansion	Formal seat in port–city planning governance with resourced technical support	Extrapolated from port–city digital-twin and community-transparency literature [12, 13]	Number of formal community consultations held per expansion decision	Lack of resourced, independent technical support for community participation

Note: IoT = Internet of Things; PM10 = particulate matter with an aerodynamic diameter of 10 µm or less.

5.4 Positioning Relative to Prior Literature

Compared with prior literature on broader urban, transport, and multimodal logistics digitalization [9, 21, 28], this review redirects attention toward the comparatively underexplored bulk segment. Within the available bulk-specific literature, dry-bulk and terminal-automation applications are better represented than liquid-bulk applications, while grain/agribulk and dust/air-quality studies are more common than research on mineral, cement, or petrochemical bulk chains. These gaps are further acknowledged in the limitations below. The review also provides an explicit mapping between major technology clusters and the economic, environmental, and social dimensions of sustainability at the port–city interface, while emphasizing the governance conditions that may influence the realization of these sustainability benefits.

6 Limitations and Future Research

6.1 Limitations

As an integrative rather than systematic review, this study relies on the published and industry literature identified through the search strategy described in Section 3 and does not claim systematic-review-level exhaustiveness. The literature sample remains modest relative to the breadth of the topic, and bulk-specific evidence remains limited compared with the broader port and logistics literature. The review may also be subject to publication bias, as successful digitalization projects are more likely to be reported than unsuccessful or abandoned initiatives. In addition, container-terminal and general-port evidence still dominates the literature relative to bulk-specific evidence, and findings extrapolated from broader port and logistics contexts to bulk applications should therefore be interpreted cautiously pending further validation.

Geographical bias toward developed ports in Europe, North America, and East Asia is also evident in the reviewed literature, while evidence from developing-city bulk terminals remains comparatively limited. The heterogeneity of dry and liquid bulk cargo further limits the transferability of findings between sub-segments, particularly for dry-bulk dust and conveyor-maintenance applications. Moreover, attributing urban sustainability outcomes directly to digital technologies remains difficult because such outcomes may also be influenced by concurrent investment, regulation, or relocation decisions. Finally, this review collected no new empirical data, and the proposed four-layer framework and propositions, although grounded in the reviewed literature, have not yet been empirically tested in a bulk cargo setting.

6.2 Future Research Agenda

Future research should pursue: (i) empirical case studies of digitalization programs in bulk terminals, disaggregated by dry versus liquid bulk, and their measured urban impacts; (ii) development and validation of bulk-specific digital twins at the port–city interface, extending the dry-bulk biomass-terminal precedent [2] to other commodity types; (iii) longitudinal assessment of environmental sensing programs around bulk facilities, testing whether PM10-based monitoring translates into measurable community-health or zoning outcomes [5]; (iv) direct empirical tests of Propositions P1–P5, particularly the governance-as-moderator proposition (P2) and the relationship between efficiency gains and rebound effects (P4); (v) governance research on data-sharing arrangements between ports, cities, and bulk supply chain actors, including who bears the cost and who controls access to shared environmental and traffic data; and (vi) comparative research on digitalization capacity and outcomes in developing-city bulk ports, where the current evidence base remains limited.

7 Conclusions

This review examined the role of digitalization and innovative technologies in bulk cargo transportation from the perspective of sustainable port–city development. Five major technology clusters were identified: IoT sensing, AI and predictive analytics, digital twins, blockchain-enabled documentation, and automation. These technologies were linked to economic, environmental, and social sustainability outcomes, and a four-layer conceptual framework comprising physical assets, data infrastructure, intelligence services, and governance was proposed. The review also highlighted key implementation risks and trade-offs and developed a stakeholder-oriented roadmap to support future digitalization initiatives in bulk cargo systems.

Data Availability

No new data were generated or analyzed in this study. All information used in the review is available from the cited sources.

Conflicts of Interest

The author declares no conflicts of interest.

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